Web Programming

Rodrigo Martins Pagliares

Last update: 10/27/2025

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JavaScript
Dynamic client-side scripting

Introduction

Introduction

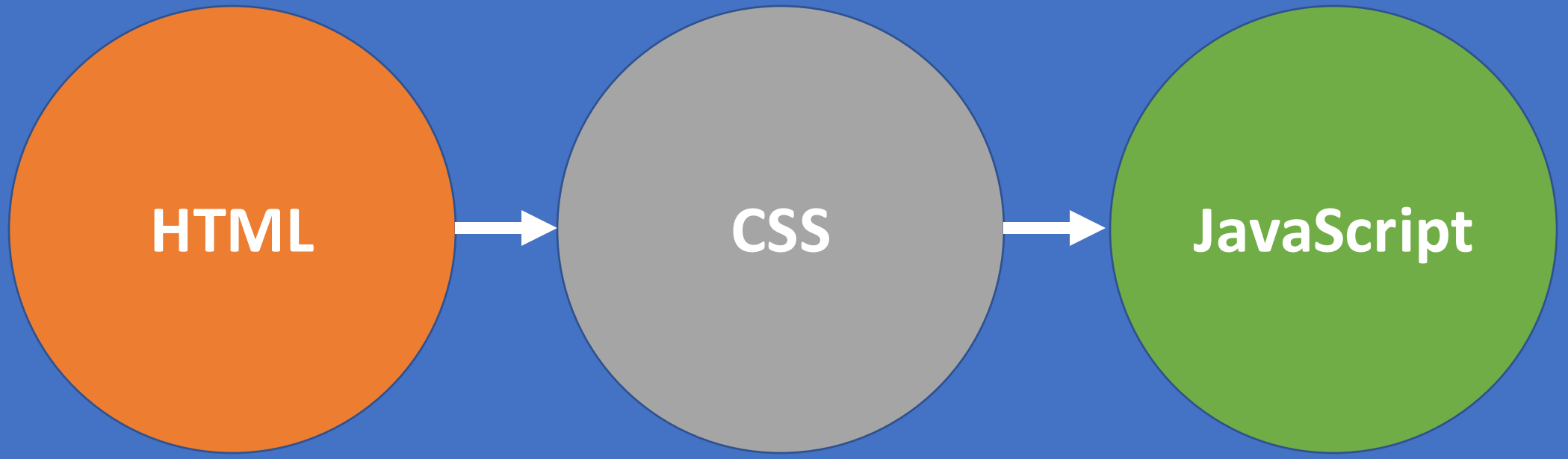
- **JavaScript** is a **programming language** that **allows** you to implement **complex things** on **web pages**.

- **Every time** a **web page** does **more** than just sit there and **display static information** for you to look at you can **bet** that **JavaScript** is probably **involved**.

What can be done with JavaScript?

- **JavaScript** allows, for **instance**:
  - ✓Display timely **content updates**
  - ✓Display **interactive maps**
  - ✓Display animated **2D/3D graphics**
  - ✓**Scroll video** jukeboxes
  - ✓and **much more**…

Learning pathway



This part of the course

Modules of this part of the course

1. JavaScript first steps
2. JavaScript building blocks
3. Introducing JavaScript objects
4. Asynchronous JavaScript
5. Client-side web APIs

Filter

HTML — Structuring the web

- Introduction to HTML
- Multimedia and embedding
- HTML tables

CSS — Styling the web

- CSS first steps
- CSS building blocks
- Styling text
- CSS layout

JavaScript — Dynamic client-side scripting

- JavaScript first steps
- JavaScript building blocks
- Introducing JavaScript objects
- Asynchronous JavaScript
- Client-side web APIs

Web forms — Working with user data

Learn web development

Welcome to the MDN learning area. This set of articles aims to guide complete beginners to web development with all that they need to start coding websites.

The aim of this area of MDN is not to take you from "beginner" to "expert" but to take you from "beginner" to "comfortable." From there, you should be able to start making your way, learning from the rest of MDN, and other intermediate to advanced resources that assume a lot of previous knowledge.

If you are a complete beginner, web development can be challenging — we will hold your hand and provide enough detail for you to feel comfortable and learn the topics properly. You should feel at home whether you are a student learning web development (on your own or as part of a class), a teacher looking for class materials, a hobbyist, or someone who just wants to understand more about how web technologies work.

Looking to become a front-end web developer?

We have put together a course that includes a
towards your goal.

Get started

In this article

Where to start
Topics covered
Getting our code examples
Contact us
See also



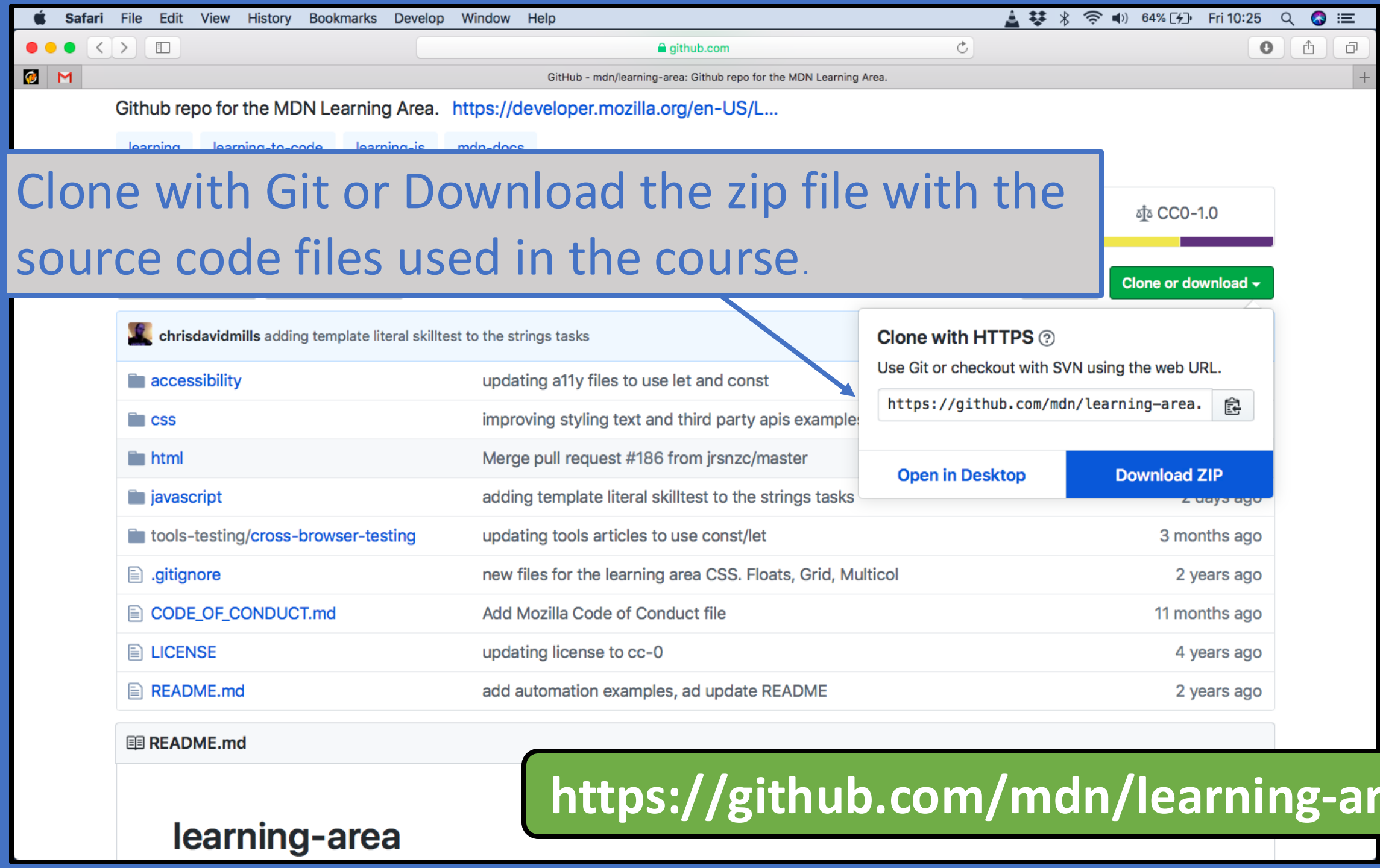
How to uplevel your hybrid cloud connection on Equinix Metal using Equinix Fabric

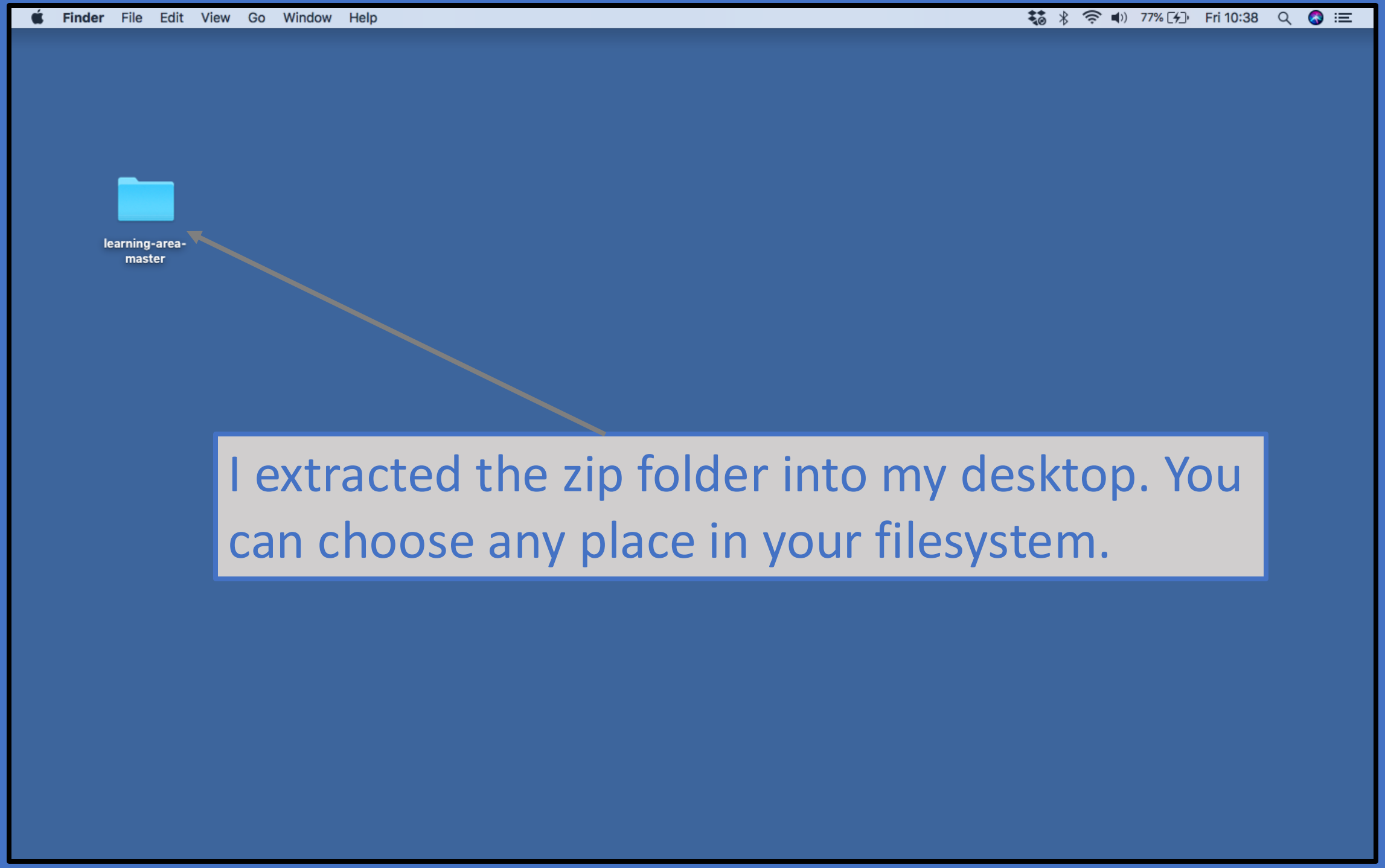
Mozilla ads

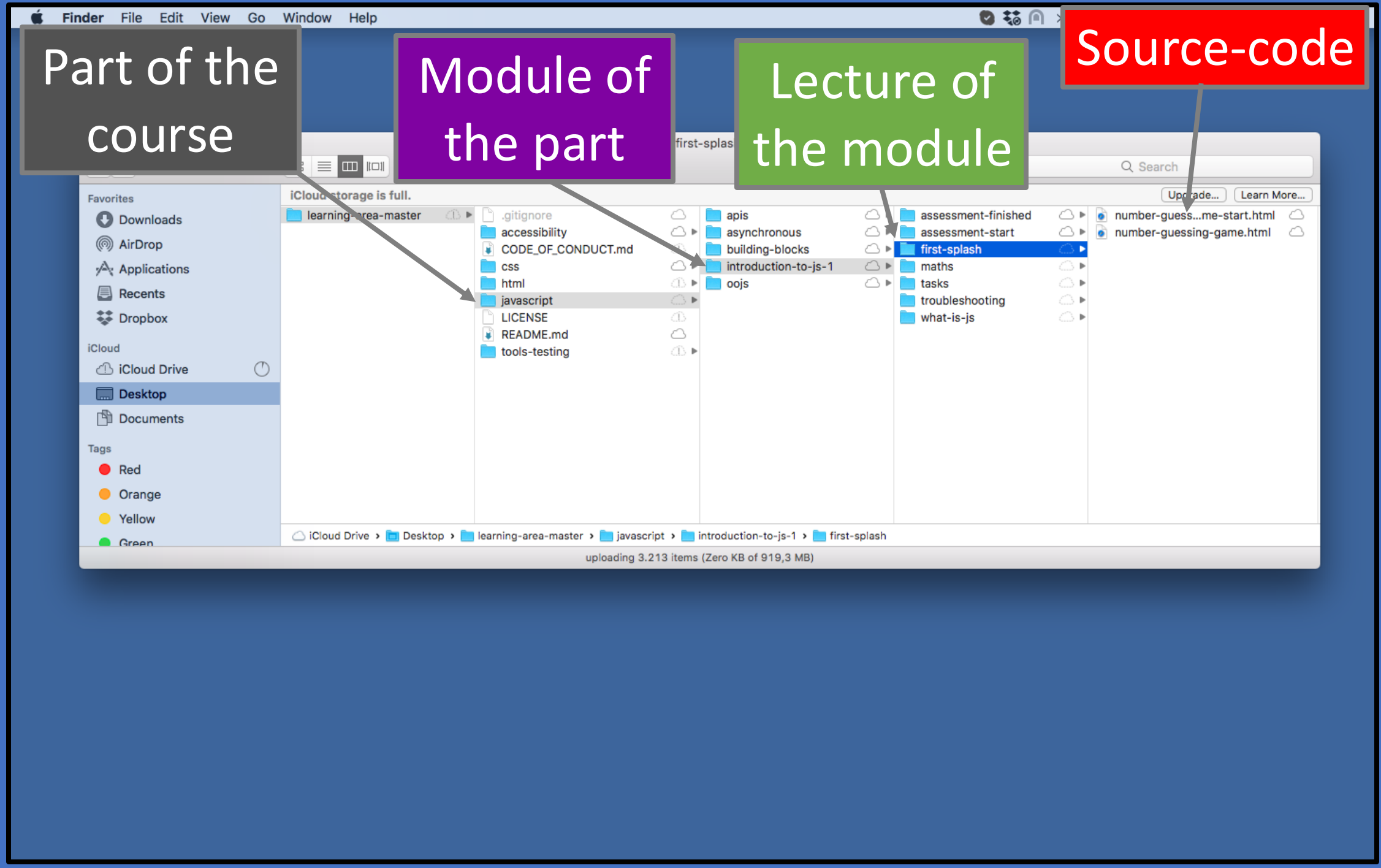
https://tinyurl.com/mryx9yzw

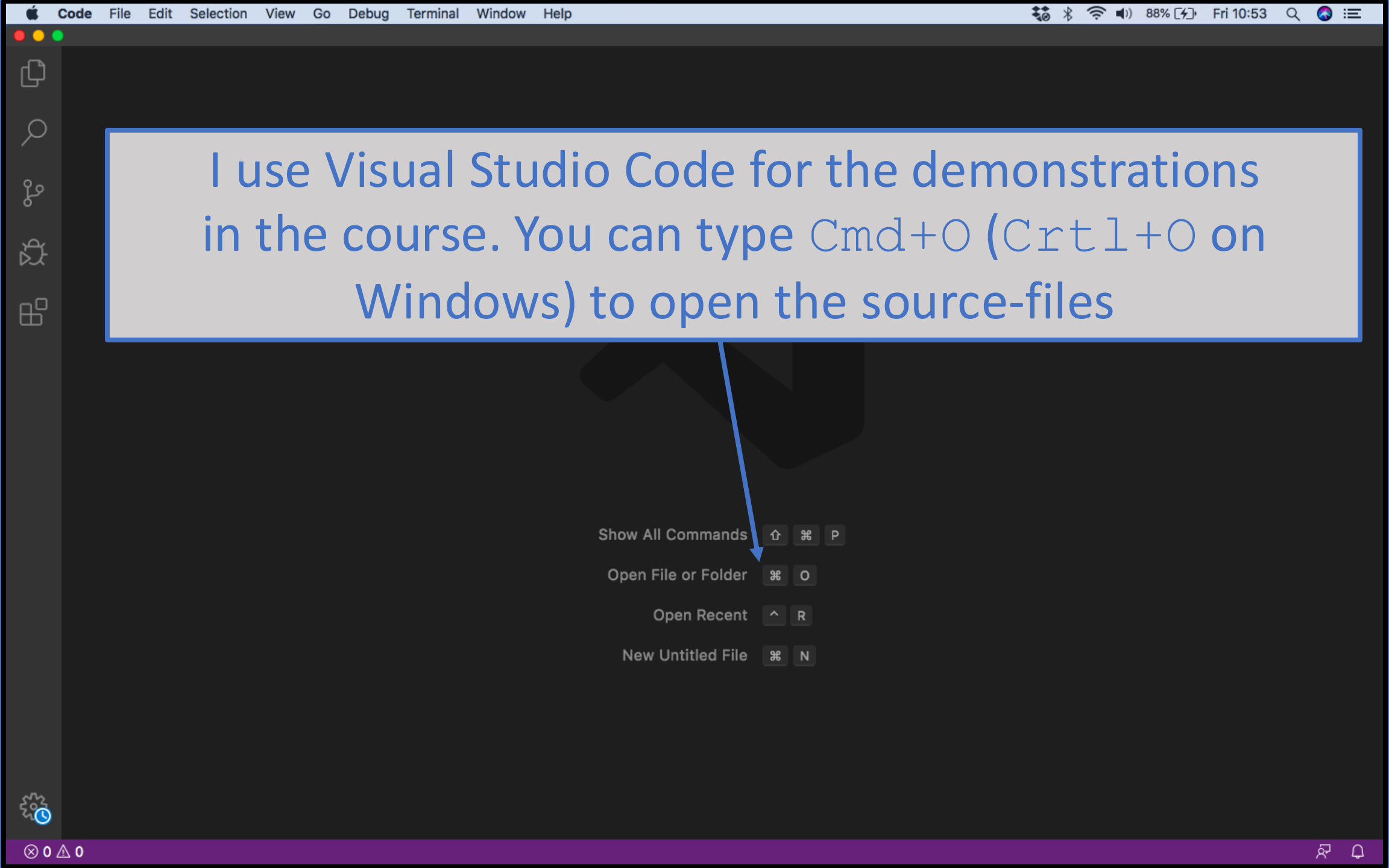
How to use the source-code files

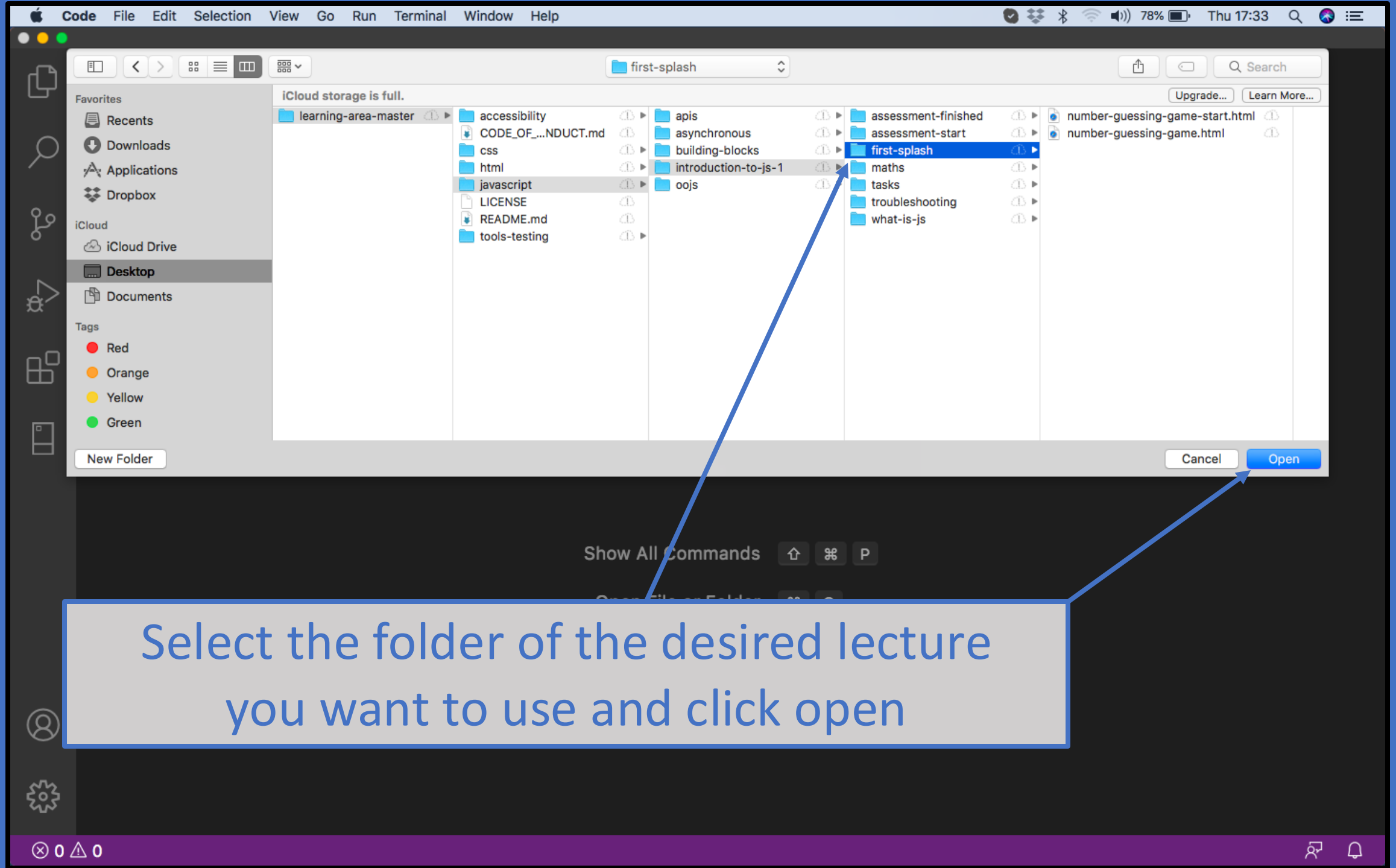
Lecture: How to use the source-code files
Part: JavaScript - Dynamic client-side scripting

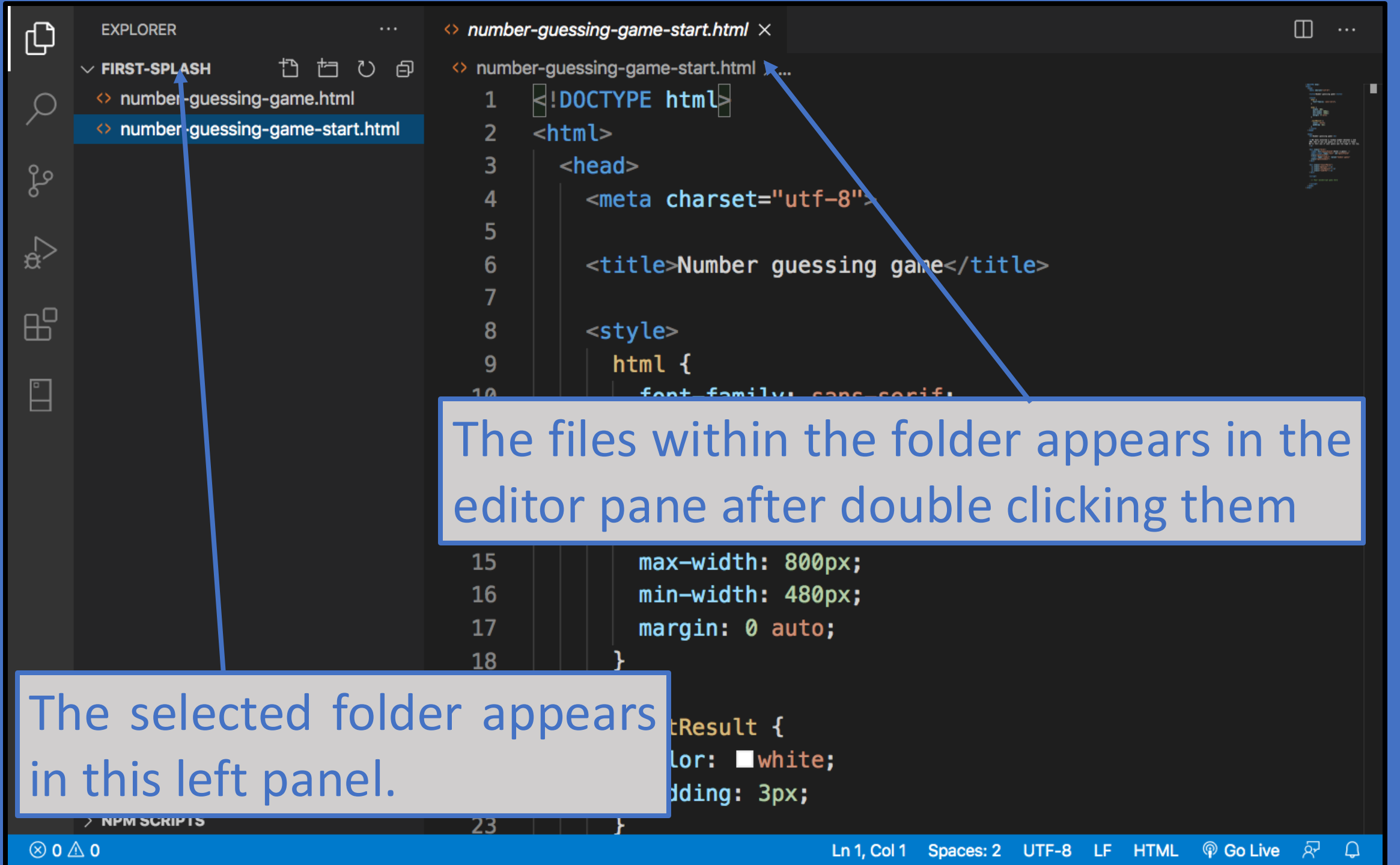

Safari File Edit View History Bookmarks Develop Window Help
64% Fri 10:25
github.com
GitHub - mdn/learning-area: Github repo for the MDN Learning Area.
Github repo for the MDN Learning Area. https://developer.mozilla.org/en-US/L...
learning learning-to-code learning-js mdn-docs
Clone with Git or Download the zip file with the source code files used in the course.
CC0-1.0
Clone or download
chrisdavidmills adding template literal skilltest to the strings tasks
Clone with HTTPS
Use Git or checkout with SVN using the web URL.
https://github.com/mdn/learning-area.
Open in Desktop
Download ZIP
accessibility updating a11y files to use let and const
css improving styling text and third party apis example
html Merge pull request #186 from jrsnzc/master
javascript adding template literal skilltest to the strings tasks
tools-testing/cross-browser-testing updating tools articles to use const/let 3 months ago
.gitignore new files for the learning area CSS. Floats, Grid, Multicol 2 years ago
CODE_OF_CONDUCT.md Add Mozilla Code of Conduct file 11 months ago
LICENSE updating license to cc-0 4 years ago
README.md add automation examples, ad update README 2 years ago
README.md
learning-area
https://github.com/mdn/learning-area

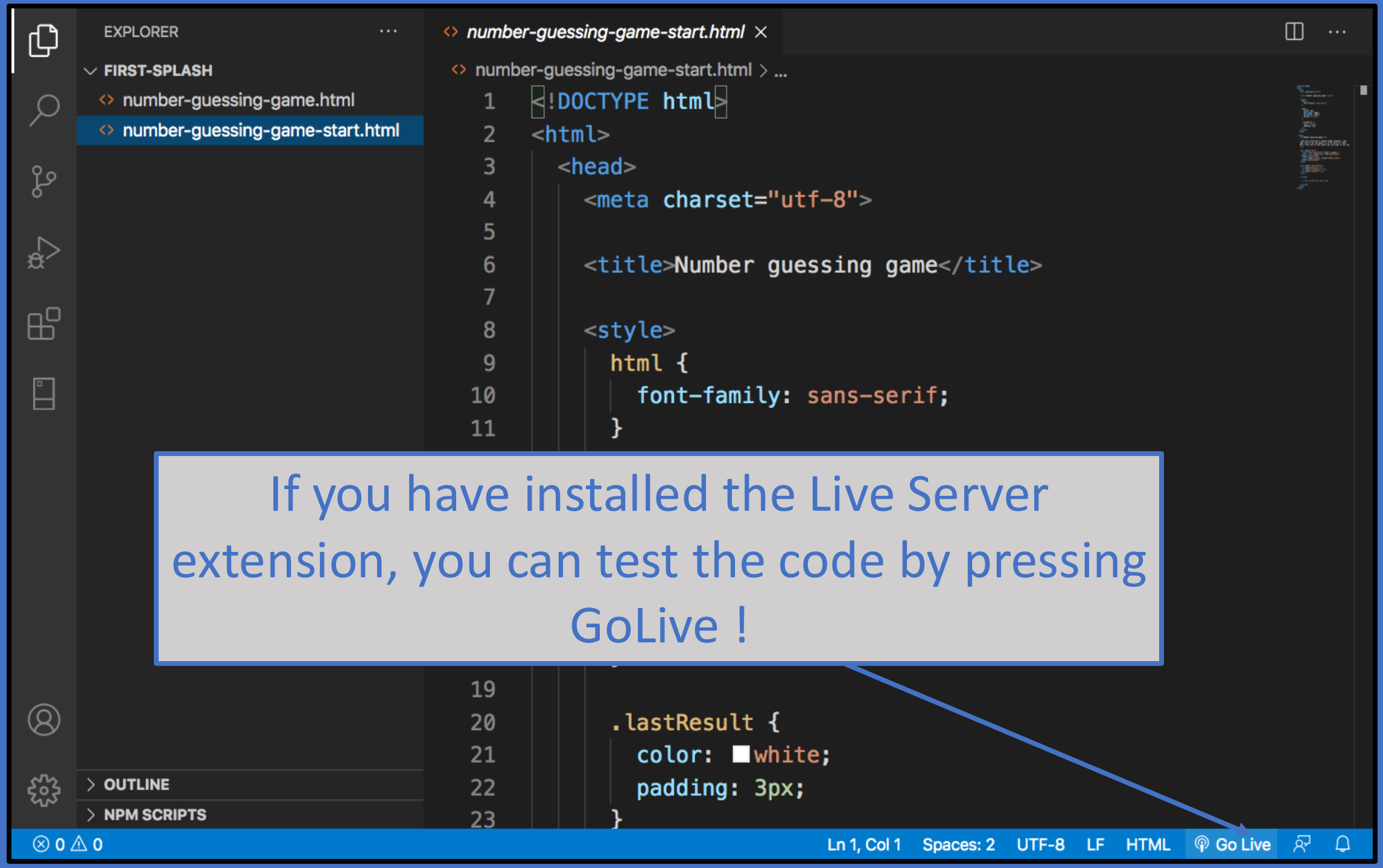

Finder File Edit View Go Window Help
77% Fri 10:38
learning-area-master
I extracted the zip folder into my desktop. You can choose any place in your filesystem.

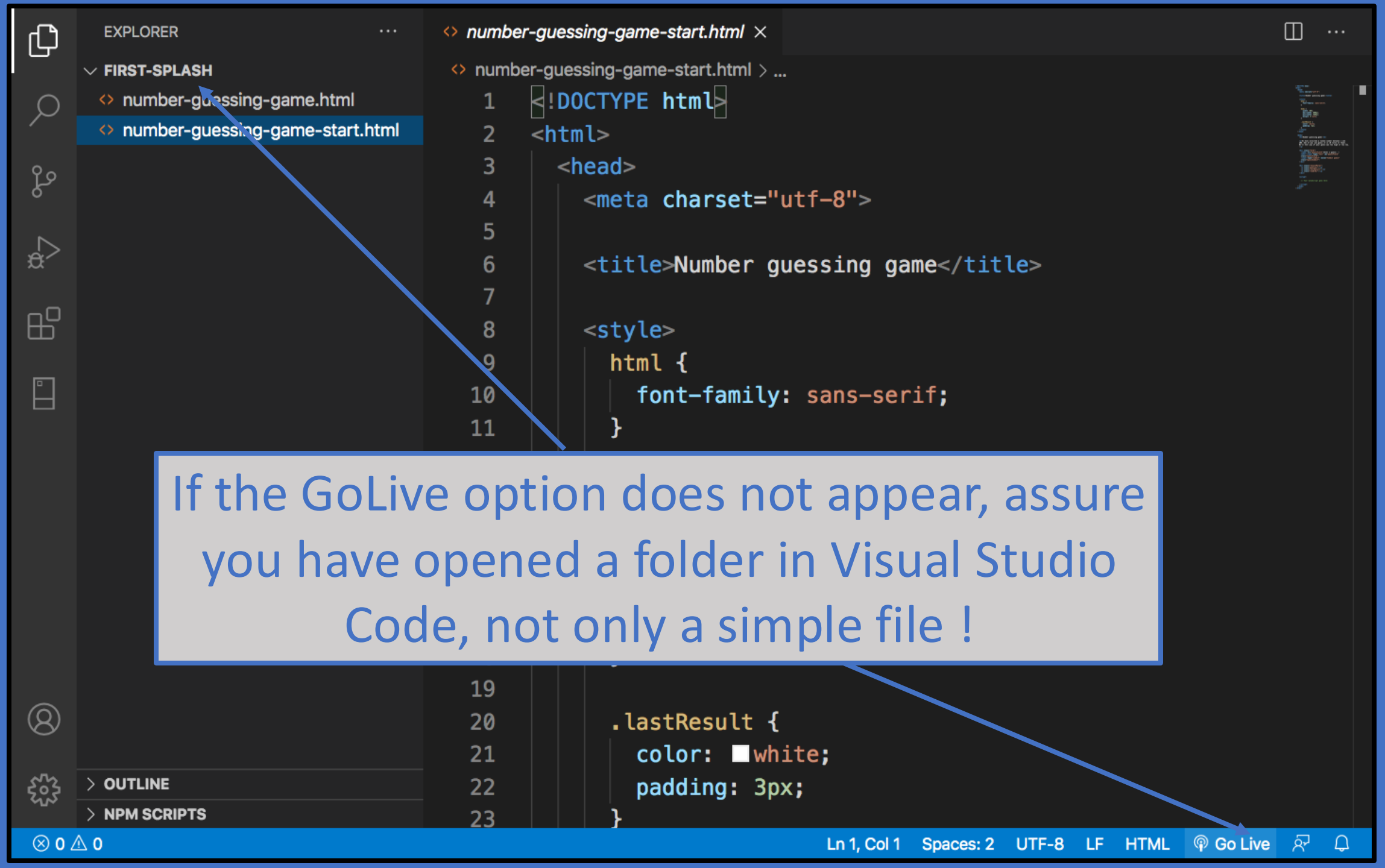

Part of the course
Module of the part
Lecture of the module
Source-code
Finder File Edit View Go Window Help
first-splas
Search
Favorites
Downloads
AirDrop
Applications
Recents
Dropbox
iCloud
iCloud Drive
Desktop
Documents
Tags
Red
Orange
Yellow
Green
iCloud storage is full.
Upgrade...
Learn More...
learning-area-master
.gitignore
accessibility
CODE_OF_CONDUCT.md
css
html
javascript
LICENSE
README.md
tools-testing
apis
asynchronous
building-blocks
introduction-to-js-1
oojs
assessment-finished
assessment-start
first-splash
maths
tasks
troubleshooting
what-is-js
number-guess...me-start.html
number-guessing-game.html
iCloud Drive › Desktop › learning-area-master › javascript › introduction-to-js-1 › first-splash
uploading 3.213 items (Zero KB of 919,3 MB)

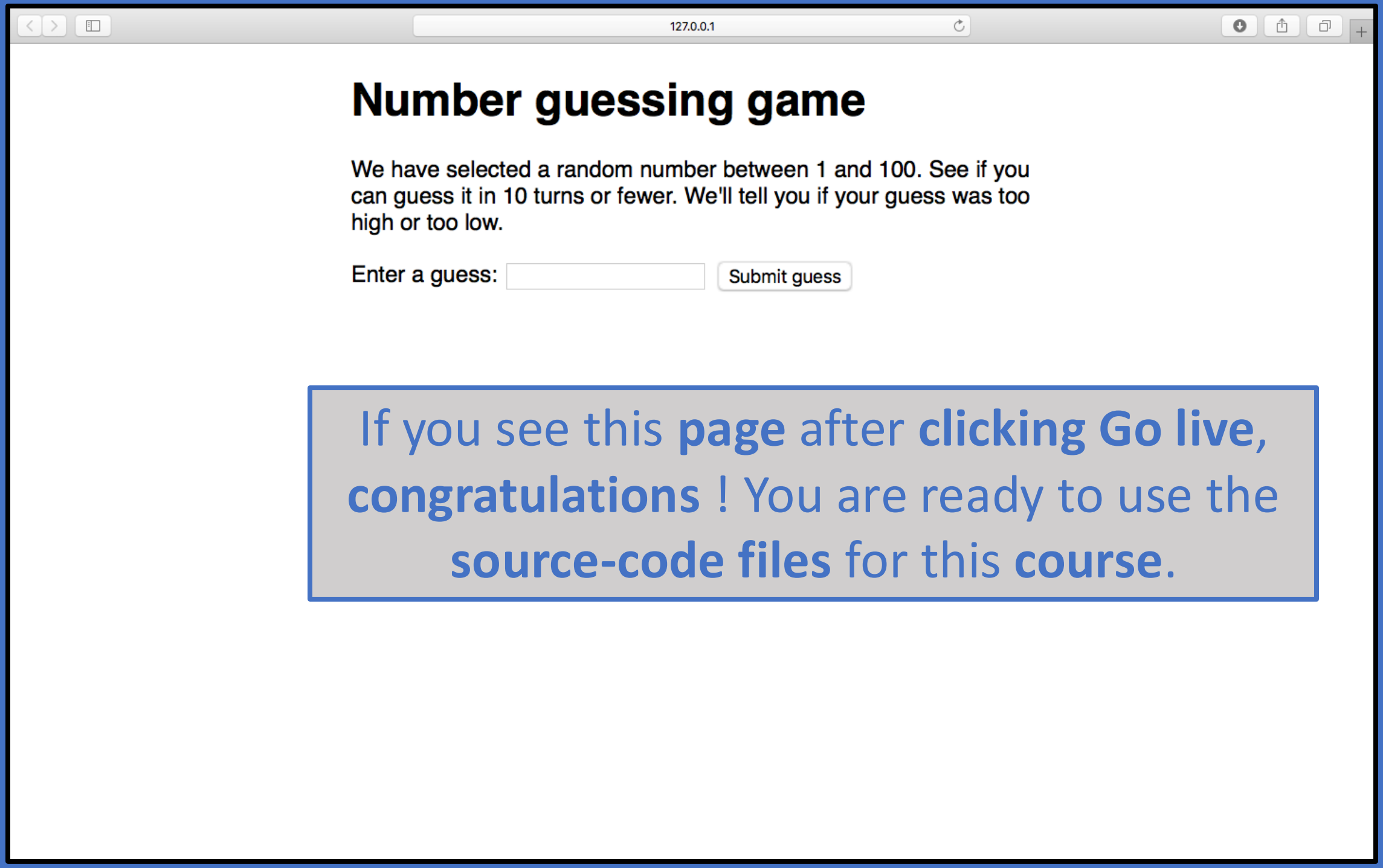

Code File Edit Selection View Go Debug Terminal Window Help
88% Fri 10:53
I use Visual Studio Code for the demonstrations in the course. You can type Cmd+O (Crtl+O on Windows) to open the source-files
Show All Commands ⇧ ⌘ P
Open File or Folder ⌘ O
Open Recent ^ R
New Untitled File ⌘ N
⊗ 0 ⚠ 0


Code File Edit Selection View Go Run Terminal Window Help
Thu 17:33
first-splash
Search
Favorites
Recents
Downloads
Applications
Dropbox
iCloud
iCloud Drive
Desktop
Documents
Tags
Red
Orange
Yellow
Green
iCloud storage is full.
Upgrade...
Learn More...
learning-area-master
accessibility
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css
html
javascript
LICENSE
README.md
tools-testing
apis
asynchronous
building-blocks
introduction-to-js-1
oojs
assessment-finished
assessment-start
first-splash
maths
tasks
troubleshooting
what-is-js
number-guessing-game-start.html
number-guessing-game.html
New Folder
Cancel
Open
Show All Commands
Select the folder of the desired lecture you want to use and click open


EXPLORER
FIRST-SPLASH
number-guessing-game.html
number-guessing-game-start.html
number-guessing-game-start.html
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <meta charset="utf-8">
6 <title>Number guessing game</title>
8 <style>
9 html {
15 max-width: 800px;
16 min-width: 480px;
17 margin: 0 auto;
18 }
The files within the folder appears in the editor pane after double clicking them
The selected folder appears in this left panel.
NPM SCRIPTS
Ln 1, Col 1
Spaces: 2
UTF-8
LF
HTML
Go Live


EXPLORER
FIRST-SPLASH
number-guessing-game.html
number-guessing-game-start.html
number-guessing-game-start.html
number-guessing-game-start.html > ...
1 <!DOCTYPE html>
2 <html>
3 <head>
4 <meta charset="utf-8">
5
6 <title>Number guessing game</title>
7
8 <style>
9 html {
10 font-family: sans-serif;
11 }
19
20 .lastResult {
21 color: white;
22 padding: 3px;
23 }
If you have installed the Live Server extension, you can test the code by pressing GoLive !
OUTLINE
NPM SCRIPTS
0 0
Ln 1, Col 1
Spaces: 2
UTF-8
LF
HTML
Go Live


EXPLORER
FIRST-SPLASH
number-guessing-game.html
number-guessing-game-start.html
number-guessing-game-start.html
number-guessing-game-start.html > ...
1 <!DOCTYPE html>
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8 <style>
9 html {
10 font-family: sans-serif;
11 }
19
20 .lastResult {
21 color: white;
22 padding: 3px;
23 }
If the GoLive option does not appear, assure you have opened a folder in Visual Studio Code, not only a simple file !
OUTLINE
NPM SCRIPTS
0 0
Ln 1, Col 1
Spaces: 2
UTF-8
LF
HTML
Go Live


127.0.0.1
Number guessing game
We have selected a random number between 1 and 100. See if you can guess it in 10 turns or fewer. We'll tell you if your guess was too high or too low.
Enter a guess:
Submit guess
If you see this page after clicking Go live, congratulations ! You are ready to use the source-code files for this course.

Questions?